Topic 34 — Climate Change (जलवायु परिवर्तन) (Advanced)

Covers syllabus (click to expand)

Climate Change | Atmospheric Studies | Global Climate Monitoring | Greenhouse Gas Emissions | Sources of Greenhouse Gases | Greenhouse Effect | Global Warming (भूमंडलीय तापन) | Carbon Cycle | Carbon Budget | Carbon Sink (कार्बन सिंक) | Carbon Credit (कार्बन क्रेडिट) | Carbon Trading | Climate Change Mitigation | Climate Change Adaptation | Climate Change Policy of India | IPCC (आईपीसीसी) Assessment Reports

Sources baked in: IPCC AR6; WMO; UNEP; Global Carbon Project; MoEFCC; NAPCC; India NDC; CCTS 2023; UPPCS PYQs **Weight:** ★★★ High — Kyoto vs Montreal; Emissions Gap=UNEP; carbon credit; IPCC≠UNFCCC **Last verified:** August 2026

Current Affairs (this topic)

- Advanced climate CA mirrors Climate Change — NDC / net zero / extreme-event science updates.
- Keep adaptation vs mitigation and Montreal≠Paris traps identical to the core climate chapter.

34.1 Climate Change

Identity: Long-term climate system change | Anthropogenic + natural | Broader than warming

- **Climate change** is a long-term change in the climate system — temperature, precipitation, extremes, sea level — lasting decades or longer.
- UNFCCC stresses human-driven change that alters atmospheric composition, beyond natural variability.
- IPCC AR6 finds human influence is the **dominant cause** of observed warming.
- Global warming is one part of climate change; monsoon shifts and glacier retreat also count.

Logic: Warming ⊂ climate change — do not treat them as identical.

34.2 Atmospheric Studies

Identity: Radiative forcing (विकिरण दबाव) | Keeling Curve | Warming troposphere (क्षोभमंडल) / cooling stratosphere (समतापमंडल)

- **Radiative forcing** (W/m^2) measures the net energy imbalance driving warming.

- The **Keeling Curve** records continuous atmospheric **CO_2** at **Mauna Loa, Hawaii**, since **1958** — rising trend is the classic graph.
- GHG warming typically **warms the troposphere** and **cools the stratosphere** — a fingerprint distinct from solar-only warming.
- Water vapour is a strong greenhouse gas but mainly a **feedback**, not the primary anthropogenic driver.

> **Logic:** Mauna Loa = Keeling CO_2 monitoring, not ozone hole (ओजोन छिद्र) location (ozone hole = Antarctica).

34.3 Global Climate Monitoring

Identity: WMO / NOAA / NASA | UNEP Emissions Gap | Global Carbon Project

- **WMO** publishes the **State of the Global Climate** report.
- **NOAA** and **NASA GISS** maintain key global temperature datasets.
- **Global Carbon Project** issues the annual **Global Carbon Budget**.
- **Emissions Gap Report** is published by **UNEP** — it measures the gap between NDCs and 1.5–2°C pathways.

PYQ — UPPCS Prelims 2024, Q6

Match: A. Human Development Report B. World Economic Outlook C. World Investment Report D. Emissions Gap Report with 1. IMF 2. UNEP 3. UNDP (यूएनडीपी) 4. UNCTAD

Options: A. 3 1 4 2 B. 3 4 1 2 C. 1 2 3 4 D. 2 3 4 1

Show answer

Ans: A — HDR=UNDP; WEO=IMF; WIR=UNCTAD; Emissions Gap=**UNEP**.

> **Logic:** Emissions Gap Report ≠ IPCC Assessment Report.

34.4 Greenhouse Gas Emissions

Identity: CO_2-led basket | Absolute vs per capita | India rank traps

- Major anthropogenic GHGs: **CO_2, CH_4, N_2O**, and fluorinated gases (HFCs, PFCs, SF_6).
- Emissions are often reported as **CO_2-equivalent (CO_2e)** using **GWP**.
- India is among the **top absolute emitters** (~**3rd** globally) but has **low per capita** emissions.
- ** **PYQ — UPPCS Prelims 2023, Q109**

What was India's rank in the world in greenhouse gas emissions in 2022?

Options: A. 9th B. 10th C. 11th D. 12th

Show answer

Ans: B — Paper answer **10th**. Absolute total emitter rank is commonly ~**3rd** — do not mix the two frames.

PYQ — UPPCS Prelims 2018, Q132

Largest CO_2 emitter in 2015:

Options: A. China B. USA C. India D. France

Show answer

Ans: A

Logic: Absolute emitter rank ≠ CCPI / "performance" rank.

34.5 Sources of Greenhouse Gases

Identity: Energy / industry / AFOLU / transport / waste | Methane hydrates | Rice CH_4+N_2O

- **Energy and electricity** (coal, oil, gas) dominate CO_2.
- **Industry** adds process CO_2 (cement, steel) plus F-gases.
- **AFOLU** (agriculture, forestry, land use): **CH_4** from livestock/rice; **N_2O** from fertilisers; CO_2 from deforestation.
- **Rice** is a major anthropogenic source of **both methane and nitrous oxide**.
- Natural **wetlands** dominate natural methane; **methane hydrates** under Arctic tundra (टुंड्रा) and seafloor can release CH_4 if warmed.
- **Transport** and **buildings** add fossil CO_2; **waste** adds landfill methane.
- Crop/biomass residue burning can release **CO, SO_2, CH_4, and ozone-related pollutants**.
- **Argon is not a greenhouse gas.**

PYQ — UPPCS Prelims 2022, Q102

Which is NOT a greenhouse gas?

Options: A. Argon B. CFC C. Methane D. Carbon dioxide

Show answer

Ans: A

Logic: Detailed gas-by-gas chemistry also appears in Topic 37. Solar flares are **not** a primary recent climate driver.

34.6 Greenhouse Effect

Identity: Fourier | Natural vs enhanced | IR trapping | –18°C without it

- **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
- The **natural greenhouse effect** (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth habitable (~**+15°C** average; without it ≈ **–18°C**).
- Shortwave solar energy reaches Earth; the surface re-emits **longwave IR**, which GHGs absorb and re-radiate.
- The **enhanced greenhouse effect** (मानवजनित संवर्धित प्रभाव) from extra anthropogenic GHGs traps more outgoing infrared radiation.
- Main natural greenhouse gases include water vapour, CO_2, CH_4, and N_2O; CFCs/HFCs are human-made.
- **Astronomical (Milankovitch) climate theory** uses eccentricity, obliquity, and precession — **solar irradiance is not** listed in that set. **Ice cores** are the classic cryogenic climate indicator.

> **Logic:** Greenhouse effect itself is not the problem — **enhanced** anthropogenic forcing is.

34.7 Global Warming

Identity: Surface temperature rise | AR6 ~1.1°C | Impacts subset

- **Global warming** means long-term rise in global mean surface temperature.
- IPCC AR6: about **1.1°C** above the **1850–1900** baseline.
- Impacts include heatwaves, glacier melt, sea-level rise, coral bleaching (प्रवाल विरंजन), and monsoon extremes (UP heat / Bundelkhand drought (सूखा) / Ganga (गंगा) floods).
- Warming is necessary but not sufficient to describe all climate-change impacts.

> **Logic:** Paris aims to hold warming well below **2°C** and pursue **1.5°C**.

34.8 Carbon Cycle

Identity: Fast vs slow cycle | Atmosphere–ocean–biosphere–lithosphere | CO_2 sources trap

- The **carbon cycle** moves carbon among atmosphere, biosphere, oceans, soils, and rocks.
- **Fast cycle:** atmosphere ↔ plants ↔ soils ↔ surface ocean (years to decades).
- **Slow cycle:** rock weathering, sediment burial, fossil carbon (millions of years).
- Burning fossil fuels transfers slow-cycle carbon into the fast atmospheric pool.
- Processes that **add CO_2** to the atmosphere include **respiration**, **decay of organic matter**, **volcanic action**, and **combustion of fossil fuels**.
- **Photosynthesis consumes** CO_2 and releases oxygen. It does **not** add carbon dioxide to the carbon cycle on Earth.

- Elemental carbon occurs as crystalline allotropes such as **graphite**, **diamond**, and **fullerene**.
- In compound form, carbon appears as CO_2, carbonates, and hydrogen carbonates.

> **Logic:** Photosynthesis removes CO_2; respiration/decay/volcanoes add it.

34.9 Carbon Budget

Identity: Cumulative CO_2 cap | 1.5°C / 2°C pathways | Net zero link

- A **carbon budget** is the cumulative CO_2 that can still be emitted while limiting warming to a chosen target.
- Smaller remaining budgets for **1.5°C** than for **2°C**.
- Net zero means residual emissions are balanced by removals — not "zero gross emissions overnight."
- India links its long-term budget pathway to **net zero by 2070**.

> **Logic:** Carbon budget ≠ annual government fiscal budget.

34.10 Carbon Sink

Identity: Forests / oceans / soils | Net absorption | Blue carbon

- A **carbon sink** absorbs more CO_2 than it releases over a period.
- Major sinks: **forests**, **oceans**, **soils**, and **mangroves/blue carbon**.
- India's NDC includes a **2.5–3 $GtCO_2e$** additional forest/tree-cover sink by 2030.
- Sinks can weaken (deforestation, ocean warming) and become sources.

> **Logic:** Sink = reservoir/net absorber; sequestration = the storage process.

34.11 Carbon Credit

Identity: 1 tCO_2e | Kyoto CDM/CERs | ≠ Green Credit 2023 | Market price

- A **carbon credit** is a tradeable unit equal to **one tonne of CO_2 equivalent (1 tCO_2e)** reduced, avoided, or removed.
- The concept originated under the **Kyoto Protocol** flexible mechanisms (especially **CDM → CERs**). Projects are implemented in **developing (Non-Annex I)** countries for Annex-I buyers — CDM is **not** Annex-I-only projects.
- Credits are **traded at market prices**, not a UNEP-fixed price.
- India's **Green Credit Programme (2023)** rewards broader eco-actions — it is **not identical** to a 1 tCO_2e carbon credit.

- Related tools: **GHG Protocol** (WRI + WBCSD) for accounting; **Common Carbon Metric** for building-operation footprints; **blue carbon** for coastal sinks.

PYQ — UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from

Options: A. Earth Summit Rio B. Kyoto Protocol C. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) D. G-8 Summit

Show answer

Ans: B

> **Logic:** Carbon credit ≠ Montreal (ozone) and ≠ Green Credit Programme.

34.12 Carbon Trading

Identity: Market for credits/allowances | Kyoto CDM | India CCTS 2023

- **Carbon trading** is buying and selling emission allowances or carbon credits.
- **Cap-and-trade** sets an emissions cap and lets the market discover the price.
- Kyoto enabled international trading via **CDM, JI, and Emissions Trading**.
- India's **Carbon Credit Trading Scheme (CCTS, 2023)** is under the **Energy Conservation Act** framework, with **BEE** as a key authority (Ministry of Power).
- **PAT** trades **ESCerts** for energy efficiency — related market tool, **not** the same as carbon credits.

> **Logic:** Credit = unit; trading = market. Paris **Article 6** governs cooperative carbon markets.

34.13 Climate Change Mitigation

Identity: Cut emissions + enhance sinks | Kyoto / Paris | Renewables

- **Mitigation** reduces GHG emissions and/or enhances sinks to limit future warming.
- Tools: renewables, efficiency, electrification, afforestation, CCS (सुरक्षा कैबिनेट समिति), carbon markets, demand-side LiFE actions.
- Key treaties: **UNFCCC 1992**, **Kyoto Protocol 1997**, **Paris Agreement (पेरिस समझौता) 2015**.
- **Montreal Protocol is ozone**, not climate mitigation.

PYQ — UPPCS Prelims 2024, Q35

Which international agreement is related to climate change mitigation?

Options: A. Montreal Protocol B. Ramsar (रामसर) Convention C. Kyoto Protocol D. CITES (साइट्स)

Show answer

Ans: C

> **Logic:** Kyoto = climate. Montreal = ODS/ozone.

34.14 Climate Change Adaptation

Identity: Adjust to impacts | NAPCC missions | Disaster/heat plans

- **Adaptation** means adjusting natural or human systems to actual or expected climate impacts.
- Examples: heat action plans, drought-resistant crops, flood forecasting, coastal protection, water efficiency.
- India's adaptation spine includes NAPCC missions on water, Himalaya (हिमालय), agriculture, and sustainable habitat (वास स्थान), plus state action plans (SAPCC).
- Adaptation does not replace mitigation — both are required.

> **Logic:** Adapt = adjust; mitigate = reduce.

34.15 Climate Change Policy of India

Identity: NAPCC 2008 | NDC 2022 | Net zero 2070 | LiFE | CCTS

- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal ministry for climate policy and UNFCCC engagement.
- **NAPCC (2008)** has **eight national missions** (solar, efficiency, water, Himalaya, Green India, agriculture, habitat, strategic knowledge). **Nuclear power is not** a listed mission.
- **Updated NDC (2022): 45%** emissions-intensity cut vs 2005 by 2030; **50%** non-fossil electricity capacity; **2.5–3 $GtCO_2e$** forest sink.
- **Net zero by 2070** announced at **COP-26 Glasgow (2021)**, with **500 GW** non-fossil capacity by 2030 as a related pledge. Net zero means emissions are **compensated by absorption/removal**, not absolute zero every emission.
- **LiFE Mission** launched **June 2022**; idea floated at **COP-26**, not COP-25 Madrid.
- Finance/institutions often tested beside policy: **GCF (Cancun 2010)**, Kyoto **CDM**, **Earth Hour (WWF)**, **GHG Protocol (WRI+WBCSD)**.
- Diplomacy vehicles: **International Solar Alliance**, **CDRI**.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) launched June 2022 (2) idea at COP-25 Madrid

Options: A. Only 2 B. Neither C. Both D. Only 1

Show answer

Ans: D — Idea = **COP-26 Glasgow**.

Year	Fact
2008	NAPCC
2015	Paris + first NDC + ISA
2021	Net zero 2070 (COP-26)
2022	Updated NDC; LiFE launch
2023	CCTS / Green Credit

> **Logic:** India net zero = **2070**, not 2050.

34.16 IPCC Assessment Reports

Identity: Science assessment | AR1→AR6 | Not a treaty body

- **IPCC** (WMO + UNEP parents) assesses climate science; it does **not** negotiate treaties or set emission targets.
- Three Working Groups: **I** physical science, **II** impacts/adaptation, **III** mitigation; plus a **Synthesis Report**.
- Cycle: **AR1 (1990)** → **AR2 (1995)** → **AR3 (2001)** → **AR4 (2007)** → **AR5 (2014)** → **AR6 (2021–23)**.
- **AR6 Synthesis Report (March 2023)** consolidates ~**1.1°C** warming and urgency of action.
- Special reports: **SR1.5 (2018)**, Ocean/Cryosphere (2019), Climate and Land (2019).
- **AR4** shared the **2007 Nobel Peace Prize** with Al Gore.

Report	Year	Fact
AR1	1990	Helped lead to UNFCCC
AR4	2007	Nobel year
AR5	2014	Informed Paris
AR6 SYR	**2023**	Latest synthesis

> **Logic:** IPCC ≠ UNFCCC ≠ UNEP Emissions Gap Report.
